Name of the Student	**K.Neeha**
Reg. No	**192425245**
GitHub Link	**https://github.com/192425245/ml0403.git**

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 1

Design and Develop Model

COURSE NAME: Deep Learning **COURSE CODE: MLA04**
SLOT :

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10% **Duration: 90 minutes**
Assessment Tool Weightage: 30% **Mark : 25**

Code of Conduct:
I K.Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

K.Neeha
Signature of the student: **Name of the student :K.Neeha**

Criteria	Problem Understanding & Requirement Analysis **(2)**	Model Architecture & Design **(2.5)**	Application of Deep Learning Techniques **(2.5)**	Model Development & Implementation Strategy **(2)**	Performance Analysis & Justification ***(1)***	**Total (10)**
Weightage	***20 %***	***30%***	***25%***	***15 %***	***10 %***	**100 %**
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Q1. CNN-Based Transfer Learning Model using ResNet for X-Ray Classification

Problem Understanding

The hospital has only **2,000 labeled X-ray images**, which is a small dataset for training a deep CNN from scratch. Therefore, **transfer learning** using a pre-trained ResNet model is the best approach.

Model Architecture

Input X-ray Image → Pre-trained ResNet50 → Fully Connected Layer → Output (Normal/Abnormal)

Layers to Freeze

- Freeze the **initial convolutional layer** and **early residual blocks**.
- These layers already contain general image features such as edges, textures, and shapes.

Layer to Replace

- Remove the original ResNet classification layer.
- Replace it with:
    - Global Average Pooling Layer
    - Dense Layer (256 neurons, ReLU)
    - Dropout Layer (0.5)
    - Output Layer (2 neurons, Softmax)

Training Strategy

1. Load pre-trained ResNet50 weights.
2. Freeze early layers.
3. Train only the newly added classification layers.
4. Fine-tune the last residual block with a small learning rate.
5. Use Adam optimizer and Cross-Entropy Loss.

Performance Analysis

- Reduces training time.

- Prevents overfitting.
- Achieves high accuracy even with limited X-ray images.

Q2. DenseNet-Based Model for Plant Disease Classification

Problem Understanding

The organization wants to classify **five plant diseases** using a limited dataset. DenseNet is suitable because it promotes feature reuse.

Model Architecture

Leaf Image → DenseNet121 → Global Average Pooling → Dense Layer → Softmax Output (5 Classes)

Dense Connections

In DenseNet, every layer receives feature maps from all previous layers.

Advantages of Dense Connections

- Feature reuse across layers.
- Better information flow.
- Reduces vanishing gradient problem.
- Requires fewer parameters.
- Improves learning with small datasets.

Training Strategy

1. Resize images.
2. Apply data augmentation.
3. Use pre-trained DenseNet121.
4. Replace final layer with 5-class Softmax layer.
5. Train classifier and fine-tune deeper layers.

Performance Analysis

DenseNet efficiently reuses learned features and improves classification accuracy for plant disease detection.

Q3. LSTM-Based Model for Student Performance Prediction

Problem Understanding

The university collects student information weekly:

- Attendance
- Assignment Marks
- Internal Assessment Marks
- Previous Exam Scores

Since data is sequential, an LSTM model is suitable.

Model Architecture

Input Sequence → LSTM Layer → Dropout → Dense Layer → Softmax Output

Working

1. Weekly records are provided as sequential inputs.
2. LSTM stores important information in memory cells.
3. Long-term academic patterns are captured.
4. Dense layer maps learned features to performance categories.

Output Categories

- High Performance
- Average Performance
- Low Performance

Training Strategy

- Normalize data.
- Use categorical cross-entropy loss.
- Train using Adam optimizer.
- Evaluate using accuracy and confusion matrix.

Performance Analysis

LSTM effectively captures student learning trends and predicts future academic performance accurately.

Q4. Bidirectional RNN Model for Sentiment Analysis

Problem Understanding

Customer review sentiment depends on both previous and future words. Therefore, a **Bidirectional RNN** is used.

Model Architecture

Input Review → Tokenization → Word Embedding → Bidirectional RNN → Dense Layer → Softmax Output

Major Processing Stages

1. Text Collection

Customer reviews are collected.

2. Tokenization

Reviews are converted into word tokens.

3. Word Embedding

Words are transformed into numerical vectors.

4. Bidirectional RNN Processing

- Forward RNN processes left-to-right.
- Backward RNN processes right-to-left.

5. Feature Combination

Forward and backward outputs are combined.

6. Dense Layer

Extracted features are transformed into sentiment scores.

7. Softmax Output

Predicts:

- Positive
- Negative
- Neutral

Performance Analysis

Bidirectional RNN captures complete sentence context and provides better sentiment classification accuracy.

Q5. Encoder–Decoder Model for English-to-Hindi Translation

Problem Understanding

The software company wants to translate English sentences into Hindi. Input and output sentence lengths may differ.

Model Architecture

English Sentence → Encoder → Context Vector → Decoder → Hindi Sentence

Encoder Role

- Reads the complete English sentence.
- Converts words into hidden representations.
- Extracts semantic meaning.

Hidden Representation (Context Vector)

- Stores the complete meaning of the input sentence.
- Transfers information from encoder to decoder.

Decoder Role

- Receives the context vector.
- Generates Hindi words one by one.
- Uses previous output and context information.

Output Layer

- Applies Softmax activation.
- Predicts the most probable Hindi word at each time step.

Training Strategy

1. Prepare English-Hindi sentence pairs.
2. Train using Teacher Forcing.
3. Use Cross-Entropy Loss.
4. Optimize using Adam optimizer.

Performance Analysis

Encoder–Decoder models effectively handle variable-length sequences and generate accurate English-to-Hindi translations.

Conclusion

The proposed deep learning models demonstrate the effective application of **Transfer Learning (ResNet), DenseNet, LSTM, Bidirectional RNN, and Encoder–Decoder architectures** for solving real-world problems. These models improve learning efficiency, reduce overfitting, support sequential data processing, understand contextual information, and enable accurate prediction and translation tasks.

Decoder Function

- Receives context vector.
- Generates Hindi words sequentially

Encoder Function

- Reads English sentence word by word.
- Converts words into hidden states.
- Produces context vector**.**

Hidden Representation (Co

Output Layer

Uses Softmax activation to select the most probable Hindi word at each step.

Final Conclusion

The designed models utilize advanced deep learning techniques such as Transfer Learning (ResNet), DenseNet, LSTM, Bidirectional RNN, and Encoder–Decoder Networks to solve real-world problems in healthcare, agriculture, education, sentiment analysis, and machine.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Problem Understanding & Requirement Analysis	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Model Architecture & Design	30%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Application of Deep Learning Techniques	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Model Development & Implementation Strategy	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Performance Analysis & Justification	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis